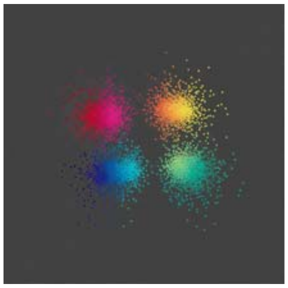
A4.3.2 Classification Techniques



A4.3.2 Explain how classification techniques in supervised learning are used to predict discretecategorical outcomes.

A4.3.2 focuses on understanding how machine learning classification techniques are utilised to predict "discrete categorical outcomes". Unlike linear regression, which predicts continuous numerical values (as covered in A4.3.1), classification aims to assign data points to specific, distinct categories or labels.

Introduction

After learning how to predict numbers with Linear Regression, we now move to another major type of supervised learning: **Classification**. Instead of asking "how much?", classification models answer the question "what kind?". They are designed to predict a **discrete categorical outcome**, or in simple terms, to assign a label to something.

What is Classification?

Classification is the process of predicting which category or class a new observation belongs to, based on patterns learned from a labelled dataset.

- **Regression Problem:** "What will the temperature be tomorrow?" (Answer is a continuous number).
- **Classification Problem:** "Will it rain tomorrow?" (Answer is a discrete category: 'Yes' or 'No').

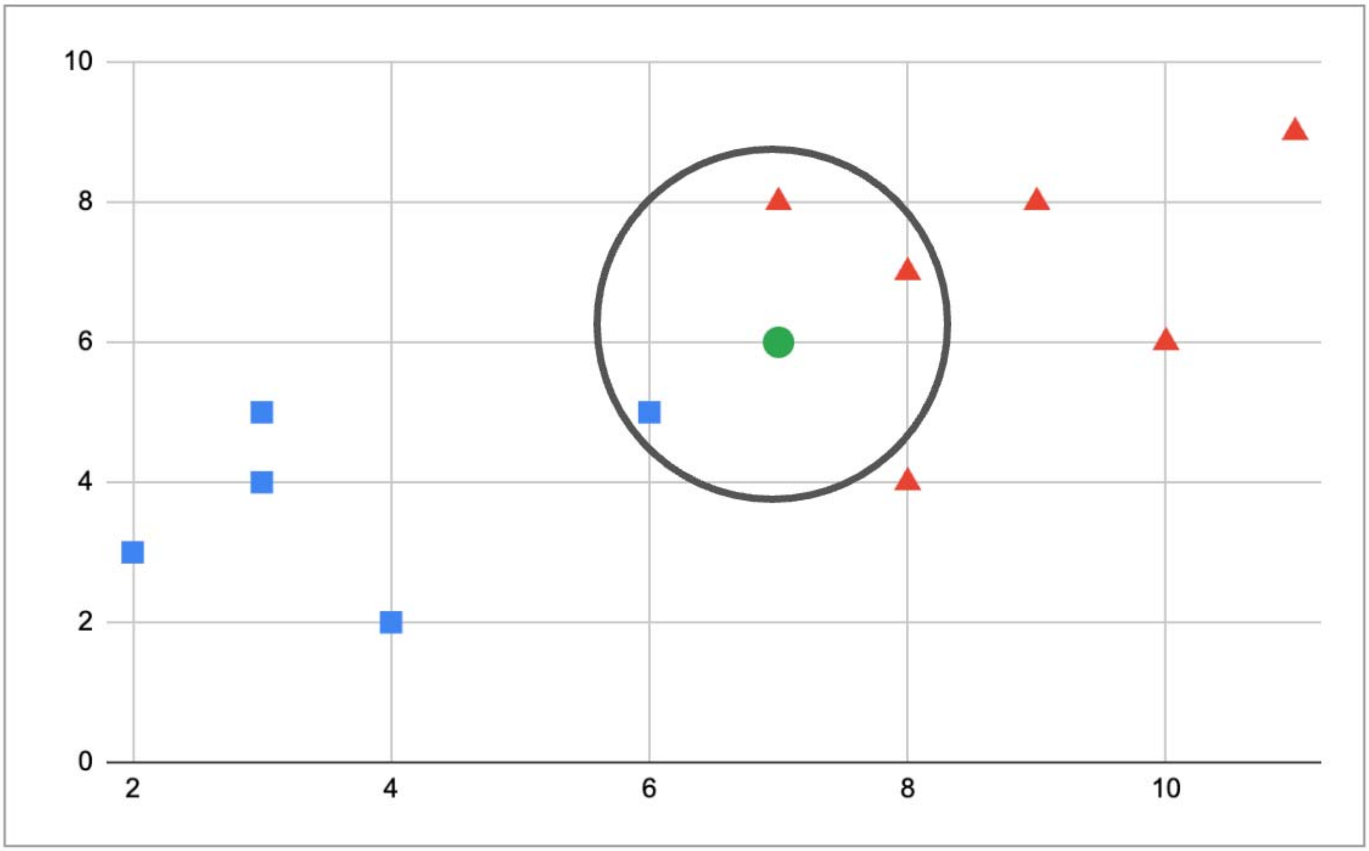
We will explore two fundamental and intuitive classification algorithms: K-Nearest Neighbours and Decision Trees.

K-Nearest Neighbors (K-NN)

The K-NN algorithm is one of the simplest and most intuitive classification methods. Its core idea can be summed up by the phrase "You are who your friends are."

- **How it Works:** To classify a new, unlabeled data point, the K-NN algorithm looks at the 'k' closest labelled data points (its "nearest neighbours") in the dataset. It then takes a **majority vote** among those neighbours. The new data point is assigned to the class that is most common among its neighbours.

In the image below ror k=3, the neighbors are 2 red triangles and 1 blue square. By majority vote, the new point is classified as a red triangle."

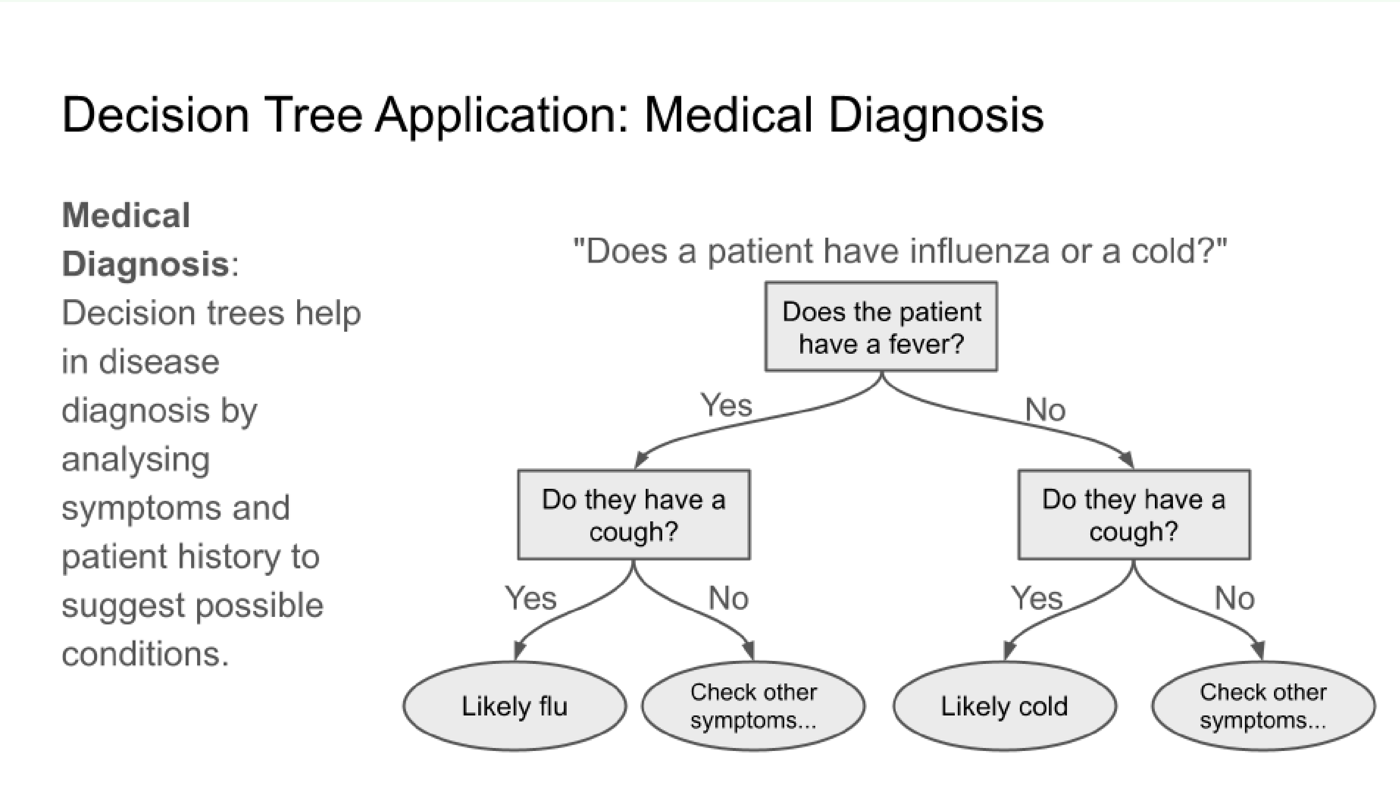


- **The Importance of 'k':** The value of 'k' (the number of neighbours to consider) is a critical **hyperparameter** that you choose. A small 'k' can be sensitive to noise, while a large 'k' can be computationally expensive.
- **Real-World Application: Collaborative Filtering Recommendation Systems** This is how services like Netflix or Amazon recommend items. To recommend a movie to you, the system finds 'k' other users who have a similar viewing history to yours (your nearest neighbours). It then looks at the movies those similar users have liked but you haven't seen yet and recommends them to you.

Decision Trees

A Decision Tree is a powerful and highly interpretable algorithm that classifies data by asking a series of if-then-else questions, much like a flowchart.

How it Works: The algorithm learns a hierarchical tree of rules from the data. To classify a new data point, you start at the **root node** and traverse down the tree, answering a question at each **decision node**. This path leads you to a **leaf node**, which gives you the final classification.



Simple Decision Tree *A simple flowchart diagram. The top box (Root Node) asks "Does the patient have a fever?". A "Yes" branch leads to another box (Decision Node) asking "Do they have a cough?". A "No" branch leads to a different decision. The paths end in oval shapes (Leaf Nodes) with the final classifications, e.g., "Likely Flu" or "Likely Cold".*

Anatomy of a Tree:

- **Root Node:** The top-most node that represents the entire dataset.
- **Decision Node:** An internal node that represents a test on a feature (e.g., "Is weight > 150g?").
- **Leaf Node:** A terminal node that represents the final class label (e.g., "Apple").

Real-World Application: Medical Diagnosis Decision trees are often used in medicine because they are highly interpretable. A doctor can look at the tree and understand exactly the logic the model used to arrive at a diagnosis. For example, a tree could guide a diagnosis based on a patient's symptoms: "If the patient has a fever AND a cough, they likely have the flu. If they have no fever BUT have a cough, they likely have a cold."

Flashcards

Use this Quizlet to learn all the key terms in this section.

Match

Ready to play?

Match all the terms with their definitions as fast as you can. Avoid wrong matches, they add extra time!

Start game

View this flashcard set

Choose a Study Mode

Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

1

The primary goal of a classification algorithm is to predict a:

A. Cumulative reward

B. Discrete categorical label

C. Hidden cluster

D. Continuous numerical value

2

The K-Nearest Neighbours (K-NN) algorithm classifies a new data point based on:

A. A linear equation.

B. A set of explicit if-then rules.

C. A majority vote of its closest neighbours.

D. The mean value of the data.

3

In a Decision Tree, what is the name for a terminal node that provides the final classification?

A. Decision Node

B. Branch Node

C. Leaf Node

D. Root Node

4

A recommendation system that suggests products to you based on what similar users have bought is an example of:

A. A Regression problem

B. Collaborative Filtering

C. Medical Diagnosis

D. A Decision Tree

5

What does the 'k' in K-NN represent?

A. The number of features in the dataset.

B. The number of classes to predict.

C. The number of times the algorithm is run.

D. The number of nearest neighbors to consider in the vote.

6

Which of these two algorithms is generally considered more "interpretable" or "white box"?

A. K-Nearest Neighbors

B. Neither is interpretable.

C. Decision Trees

D. Both are equally interpretable.

7

A Decision Tree classifies a new data point by:

A. Following a path of if-then-else questions from a root node to a leaf node

B. Evolving a population of solutions over time.

C. Finding the average of the nearest neighbours.

D. Calculating its distance to all other points.

8

Predicting whether an email is 'Spam' or 'Not Spam' is a:

A. Reinforcement Learning task.

B. Clustering task.

C. Regression task.

D. Classification task.

9

In a Decision Tree for medical diagnosis, a node that asks "Is the patient's temperature > 38°C?" is a:

A. Leaf Node

B. Root Node

C. Decision Node

D. It could be a Root or Decision Node.

10

K-NN is described as a distance-based algorithm, while a Decision Tree is described as a:

A. Layer-based algorithm

B. Reward-based algorithm

C. Rule-based algorithm

D. Population-based algorithm

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